DS	**Session**	Hashing	**Question Information**	Level 1 Challenge 93

Question description

Alice likes festivals a lot as we all do. He also likes spending money on these festivals. He spends money in buying various things on these festivals. But he has problem of forgetting. He only remembers his top three maximum spendings for any festival.
For eg, on Holi he spends 25 units on colors, 50 units on water sprays, 100 units on gifts, 150 units on sweets but he remebers only his top 3 spendings ,i.e., 50, 100 & 150.
Now as the year ends he wants to know the **festival on which he spent most of his money that he can remember**.

Constraints

$1 \leq T \leq 100$

$1 \leq N \leq 10^4$

$1 \leq X \leq 10^9$

$1 \leq |S| \leq 10; |S| \; denotes \; string \; length$

S will consist of **lowercase and/or uppercase English aplhabets**

Input Format

First line contains an integer **T**, denoting number of test cases.
Second line contains an integer **N**, denoting number of spendings in a year.
Each of the next **N** lines contain a string **S** and an integer **X** denoting festival name and spending on one of the events of that festival.

Output Format

For each test case, print a single line containing the festival name and its total spending that he remembers.
If there are more than 1 festival with the maximum spending, print the one which has lexiographically smallest name.

Logical Test Cases

Test Case 1

INPUT (STDIN)

```
2
6
B 20
A 2
A 10
A 10
B 30
A 30
3
abc 10
xyz 15
oop 8
```

EXPECTED OUTPUT

```
A 50
xyz 15
```

Test Case 2

INPUT (STDIN)

```
2
5
BA 25
AD 24
Ag 10
AE 15
AT 30
3
abc 15
xyz 25
oops 8
```

EXPECTED OUTPUT

```
AT 30
xyz 25
```

Mandatory Test Cases

Test Case 1

KEYWORD

```
while(t--)
```

Complexity Test Cases

Test Case 1

CYCLOMATIC COMPLEXITY

```
7
```

Test Case 2

TOKEN COUNT

```
260
```

Test Case 3

NLOC

```
46
```

Code

festival_spendings.c

```c
#include <stdio.h>
#include <stdlib.h>
#include <string.h>

typedef struct {
    char name[15];
    long long top[3];
    int count;
} Festival;

Festival fests[1005];
int fest_count = 0;

int find_festival(const char* name) {
    for (int i = 0; i < fest_count; i++) {
        if (strcmp(fests[i].name, name) == 0) return i;
    }
    strcpy(fests[fest_count].name, name);
    fests[fest_count].top[0] = fests[fest_count].top[1] = fests[fest_count].top[2] = 0;
    fests[fest_count].count = 0;
    return fest_count++;
}

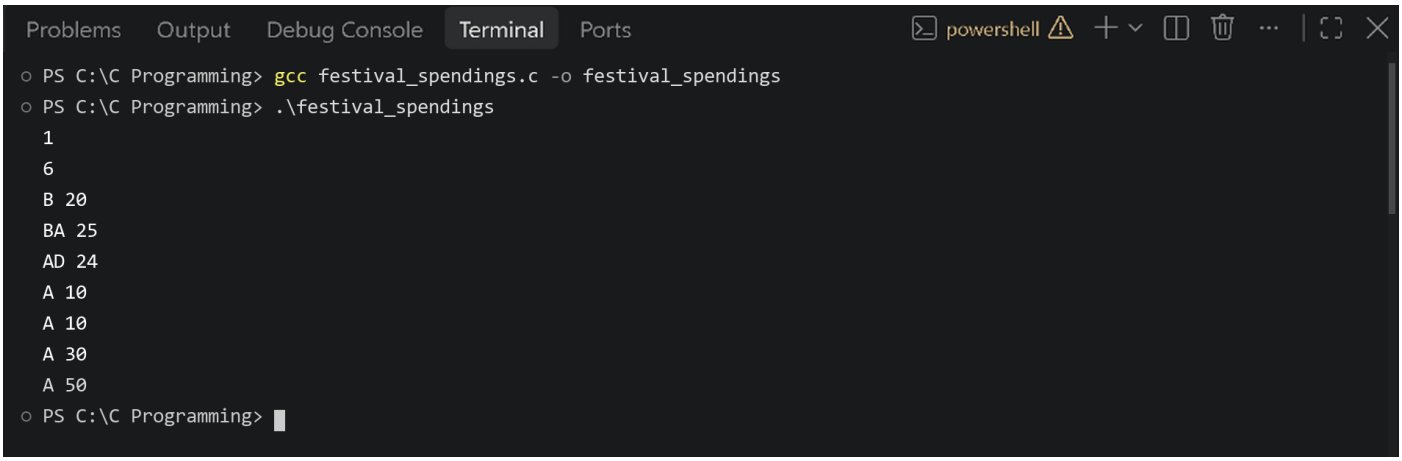
void add_spending(int idx, long long x) {
    if (x > fests[idx].top[0]) {
        fests[idx].top[2] = fests[idx].top[1];
        fests[idx].top[1] = fests[idx].top[0];
        fests[idx].top[0] = x;
    } else if (x > fests[idx].top[1]) {
        fests[idx].top[2] = fests[idx].top[1];
        fests[idx].top[1] = x;
    } else if (x > fests[idx].top[2]) {
        fests[idx].top[2] = x;
    }
    fests[idx].count++;
}

int main() {
    int t;
    if (scanf("%d", &t) != 1) return 0;

    while(t--) {
        int n;
        scanf("%d", &n);
        fest_count = 0;
        for (int i = 0; i < n; i++) {
            char name[15];
            long long x;
            scanf("%s %lld", name, &x);
            int idx = find_festival(name);
            add_spending(idx, x);
        }

        int best_idx = 0;
        long long best_sum = -1;
        for (int i = 0; i < fest_count; i++) {
            long long s = fests[i].top[0] + fests[i].top[1] + fests[i].top[2];
            if (s > best_sum || (s == best_sum && strcmp(fests[i].name, fests[best_idx].name) < 0)) {
                best_sum = s;
                best_idx = i;
            }
        }
        printf("%s %lld\n", fests[best_idx].name, best_sum);
    }
    return 0;
}
```

Output


Problems
Output
Debug Console
Terminal
Ports
powershell
PS C:\C Programming> gcc festival_spendings.c -o festival_spendings
PS C:\C Programming> .\festival_spendings
1
6
B 20
BA 25
AD 24
A 10
A 10
A 30
A 50
PS C:\C Programming>